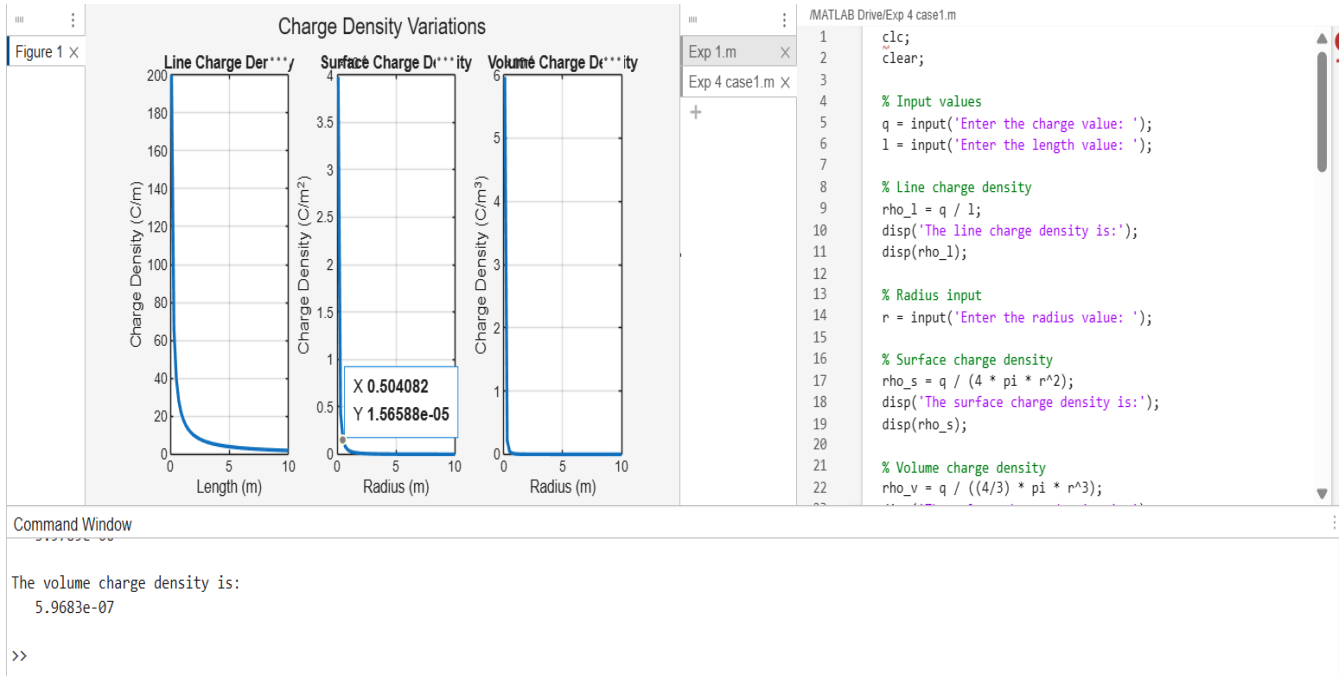
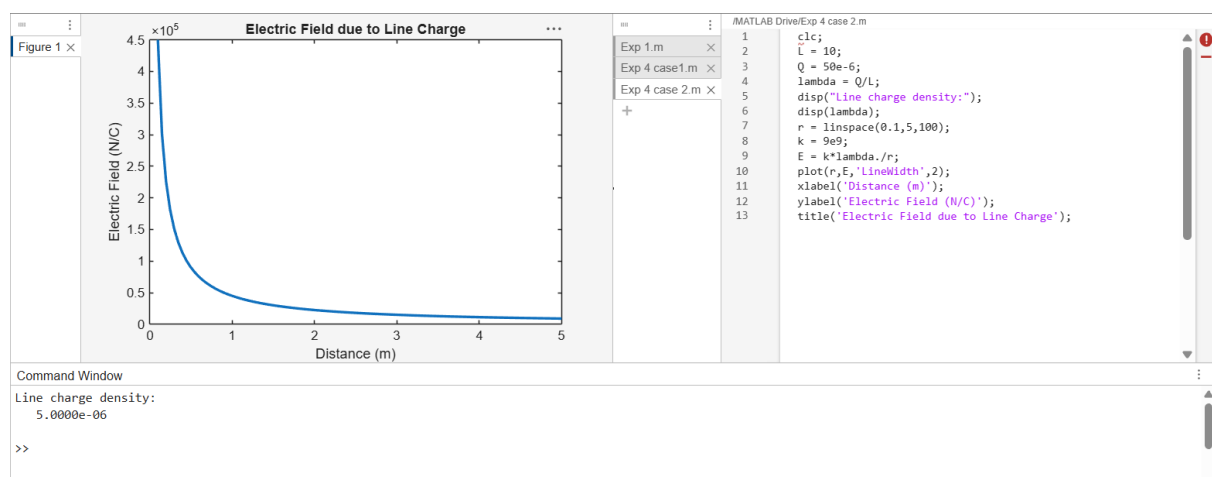


Exp. No: 4 Analysis of Continuous Charge Distribution in electric field.

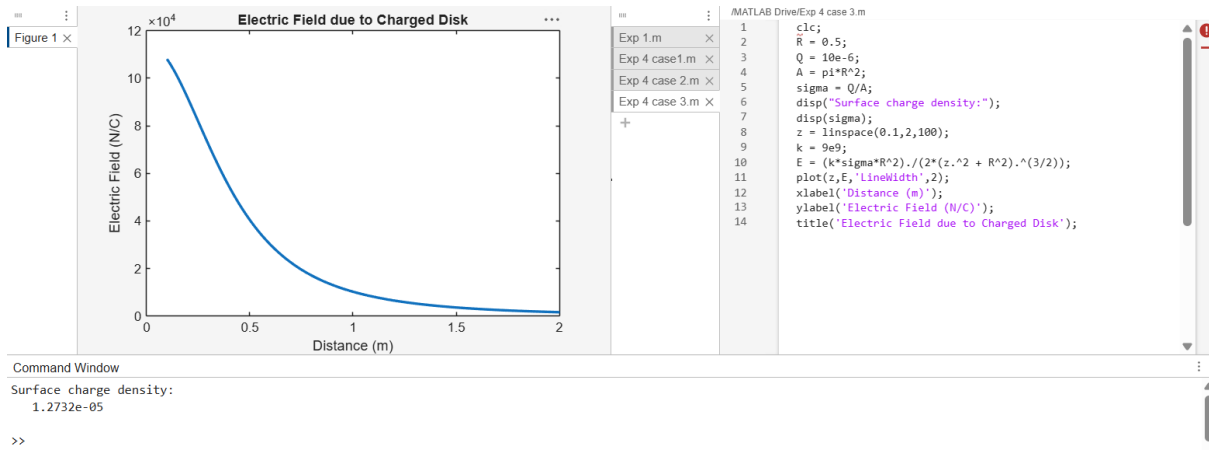
Test case (i) Determination of line, surface and volume charge density for given charge, length and radius



Test case (ii) A long wire of length $L = 10$ m carries a uniform charge of $Q = 50 \mu\text{C}$. Calculate the line charge density and plot electric field variation



Test case (iii) A uniformly charged disk of radius $R = 0.5$ m with total charge $Q = 10 \mu\text{C}$. Calculate surface charge density and electric field



Test case (iv) A uniformly charged sphere of radius $R = 2$ m with total charge $Q = 20 \mu\text{C}$. Calculate volume charge density and electric field inside and outside

